MIL-C-55532A(EL)
21 October 1969
SUPERSEDING
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MILITARY SPECIFICATION

CHARGER, BATTERY PP-1660( )/G

1. SCOPE

1.1 Scope. - This specification covers one type of portable, self-contained battery charger for lead-acid type storage batteries, designed to operate from 115 volt, 60 hertz, single phase power source, with an output of 6, 12 and 24 volts; designated Charger, Battery PP-1660( )/G. (See 6.3)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-C-596	Connector, Plug, Electrical, Connector, Receptacle, Electrical
QQ-S-571	Solder, Tin Alloy, Lead Tin Alloy and Lead Alloy
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade) and Cut Shapes
PPP-S-760	Strapping, Nonmetallic (and connectors)
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronics and Associated Electrical Equipment
MIL-V-173	Varnish, Moisture-and-Fungus-Resistant (For the treatment of Communications, Electronic, and associated Electrical Equipment)

FSC 6130

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MIL-C-3432	Cable and Wire, Electrical (Power and Control, Flexible and Extra Flexible, 300 and 600 Volts)
MIL-B-7883	Brazing of steels, copper, copper alloys, nickel alloys, aluminum and aluminum alloys
MIL-P-8585	Primer coating, zinc chromate, low-moisture-sensitivity
MIL-M-10304	Meter, Electrical Indicating, Panel Type, Ruggedized, General Specification for
MIL-M-13231	Marking of Electronic Items
MIL-C-13486	Cable, Special Purpose, Electrical: Low-Tension, Heavy Duty Single-Conductor and Multiconductor
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-F-14256	Flux, Soldering, Liquid (Rosin Base)
MIL-B-43291	Box, Fiberboard, Corrugated, Double-Wall, Weather Resistant

STANDARDS

MILITARY

MIL-STD-105	Sampling procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and containerized Unit Loads 40 inch x 48 inch Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electro-magnetic Interference of Characteristics Requirements for Equipment
MIL-STD-645	Dip Brazing of Aluminum Alloys
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester
SC-DL-357291	Charger, Battery PP-1660( )/G

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS.

* 3.1 Description. - Charger, Battery, PP-1660( )/G is a single, self-contained, full-wave, selenium rectifier battery charger. It is equipped with a forced draft cooling system, timer, fast and slow charge systems, charge rate selector switch, voltage selector switch and a metal carrying case with handle. It shall operate from 105 to 125 volt, single phase, 60 hertz source.

* 3.2 Preproduction samples. - The contractor shall furnish three preproduction samples for approval, if required by the invitation for bids and contract. (See 4.3)

3.3 Construction. - The equipment shall be constructed in accordance with Drawing SC-DL-357291.

3.4 Cables. -

3.4.1 AC power cable. - The AC power cable shall conform to Specification MIL-C-3432, type CO-03 HLF (3/14) 0565. The cable shall be 15 feet ± 6 inches in length with one end permanently attached to the charger. The free end shall be equipped with Connector, Plug, Electrical (Power, Three-wire, Grounding type) Style D22 conforming to Federal Specification W-C-596.

3.4.2 DC output cable. - Charger, Battery, PP-1660( )/G shall be equipped with two (2) 10 foot ± 6 inches output cable conforming to Specification MIL-C-13486, part number M13486/1-11. Each cable shall be permanently attached to the charger and the free ends shall be equipped with plug type ends suitable for connecting directly into the type AN, SIZE "O" military slave connections of the battery circuit of Military vehicles with 24 volt electrical systems. The positive plug shall be identified by a red colored sleeve and the negative plug by a black colored sleeve.

3.4.3 Clips, electrical, jaw-grip. - Clips, electrical, jaw-grip, into which the plug type ends of the cables shall fit, shall be furnished. The clips shall be rated at not less than 75 amperes and shall be coated with a corrosion resisting compound and permanently marked for positive and negative identification.

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3.5 Carrying case. - The charger shall be housed in a carrying case equipped with a carrying handle. Ventilating grilles for cooling air intake and exhaust shall be provided and shall be located so that the flow of cooling air will not be obstructed when the charger is operated while being located flush against a wall. Brackets shall be provided for stowing the AC power and DC output cables on the outside of the case. Two posts shall be provided for storing the two battery clips. (See 3.4.3)

3.6 Rectifier. - The rectifier shall consist of not less than 480 square inches of selenium plate area, fan cooled (see 3.11), with a temperature rise not exceeding 54°F when the equipment is operated continuously under full load in an ambient temperature of 77°F. (See 4.12)

3.7 Transformer. - The transformer shall be varnished and baked. Its temperature rise shall not exceed 117°F. when the equipment is operated continuously in an ambient temperature of 77°F. at any of the following loads with an input voltage of 115 volts ac:

Charging Current DC Amperes	Charging Voltage DC Volts
75	8
75	16
38	32

3.8 Timer. - The timer shall be spring operated and shall have a nominal rating of one hour. It shall be graduated in minutes and shall be marked every 10 minutes starting at 5 minutes. The timer shall automatically switch from fast charge rate to slow charge rate at the end of any preset time interval up to one hour. (See 4.15)

3.9 Ammeter. - The ammeter shall be in accordance with Specification MIL-M-10304, type MR36W100SPECR except scale shall be in accordance with 3.9.1, Meter Bezel shall be black and external zero corrector on front of meter shall be dull black phenolic.

3.9.1 Scale. - The scale shall contain two colored bands, one band for use with 6 and 12 volt batteries and the other for 24 volt batteries. Each band shall be colored yellow for slow charge range, green for normal fast charge and red for excessively high rate charge. The red band for 6 and 12 volt batteries shall begin at 75 amperes and for 24 volt batteries at 38 amperes.

3.10 Slow-charge system. - Slow battery-charging rates shall be obtained by reducing the primary voltage of the transformer by means of a reactor connected across the timer switch terminals. The reactor shall be protected against overload by an automatic-reset thermal-overload circuit breaker.

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3.11 Cooling system. - The battery charger shall have a forced ventilation cooling system utilizing a motor-driven fan. The temperature rise of the fan motor shall not exceed 117°F. when the charger is operating at maximum output in an ambient temperature of 125°F. Motor shall be fungus treated, rust proofed, Class B insulation for high temperature operation and special lubricant, Andreol or equal for satisfactory operation at high and low temperature. The fan shall be mounted directly on the motor shaft. (See 4.12)

3.12 Circuit breakers. - Circuit breakers shall be provided in the AC input circuit for protection of the transformer and rectifier and in the DC charging circuit for protection against damage as result of connection to dead or shorted batteries, incorrect connections to the batteries being charged, and shorting of the charging leads. The circuit breaker shall be of the automatic reset type. A protective relay shall be provided to prevent inverse voltage damage to the selenium rectifier.

3.13 Controls. - The controls shall be located on or in the carrying case, in protected locations and shall be easily accessible and visible when charger is in use at floor level or bench level. The controls shall be provided with knobs designed for easy manipulation by an operator wearing heavy mittens or gloves.

3.13.1 Charge rate switch. - The charge rate switch shall be marked "off", "low", "med" and "high".

3.13.2 Battery voltage selector switch. - The battery voltage selector switch shall have 6, 12, and 24 volt positions.

3.13.3 On-off switch. - An "on-off" switch shall be provided for the AC power supply.

3.14 Size and weight. - The equipment shall be constructed in accordance with drawings which controls the size and weight requirements.

3.15 Cleaning. -

* 3.15.1 Parts. - After fabrication, parts shall be cleaned in accordance with Requirement 9 of MIL-STD-454.

3.15.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, such cleaning shall also be performed before final assembly of the equipment.

3.16 Finish. - The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4)

3.17 Marking. -

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3.17.1 General. - Marking shall conform to Specification MIL-M-13231. (See 4.4) Front panel marking shall be group I as described in that specification.

3.17.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.17.3 Serial numbers. - Charger, Battery PP-1660( )/G shall have serial numbers.

* 3.18 Riveting. - Rivets shall conform to requirement 12 of MIL-STD-454. Rivets shall be wet driven using a zinc-chromate primer conforming to MIL-P-8585. For aluminum, the distance from the centers of the rivet holes to the edges of the material in which the rivets are placed shall be not less than 2 times the rivet diameter and 2-1/2 times for magnesium. The size and spacing of rivets on aluminum plates are sometimes dictated by factors other than those mentioned above. The following are some general rules that may be helpful in proportioning aluminum riveted joints and connections:

(a) The diameter of a rivet should be not larger than 3 times the thickess of the outside plates or shapes, and should be not smaller than the thickness of the thickest plate in which the rivet is driven.

(b) A rivet used in the leg of an angle should have a diameter not exceeding 0.3 the width of the leg.

(c) The minimum spacing of rivets should be 3 times the nominal rivet diameter. The maximum spacing is often governed by the possibility of compression buckling of parts between rivets. Where buckling or strength requirements do not control, the spacing should still be held below some reasonable maximum limit, such as 24 times the thickness of the sheet or plate.

(d) In riveted joints that must be made pressure tight by caulking the edge of the plates, the edge distance should not exceed 1 and 1/2 times the diameter or 4 times the thickness of the plate. Spacing of rivets should not exceed 4 times the rivet diameter nor 10 times the thickness of the plate.

* 3.19 Soldering and brazing. -

* 3.19.1 Brazing. - Brazing of steels, copper, copper alloys, and nickel alloys shall conform to MIL-B-7883. Brazing of aluminum alloys shall conform to MIL-STD-645.

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* 3.19.2 Soldering. - Soldering shall conform to requirement 5 of MIL-STD-454. Solid solder shall conform to type S and fluxcored solder to type R or RMA of QQ-S-571. Liquid rosin flux for use with solid solder shall conform to MIL-F-14256. For soldering electrical connections, composition SN60 or SN63 shall be used unless another composition is required for a specific application.

* 3.19.3 Rosin-base residues. - Flux residues shall be removed by applying organic solvents. Mechanical means such as agitation, brushing, etc, may be used in conjunction with the solvents. The cleaning solvents and methods used shall not have any deleterious effect on the parts, materials, and assemblies being cleaned.

* 3.20 Tropicalization of material and parts. - Fungus-inert materials shall be in accordance with requirement 4 of MIL-STD-454, and shall be used where practicable. When materials are required to be treated, they shall be cleaned in accordance with requirement 9 of MIL-STD-454 and then shall be treated in accordance with 3.20.1 through 3.20.6.

* 3.20.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

* 3.20.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the items in which the treated material is used, or during use of the finished items when used for the purpose intended.

* 3.20.3 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

* 3.20.4 Statement of treatment. - The statement of treatment shall indicate whether or not nutrient materials have been used. If such materials have been used, it shall also describe in detail the materials to be treated and the treating materials and processes that the contractor proposes to use.

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* 3.20.5 Overall treatment (moisture and fungus proofing treatment (MFP)). - When overall tropicalization of equipment is necessary, a moisture and fungus resistant coating of varnish conforming to MIL-V-173 shall be applied in accordance with MIL-T-152. The moisture-fungus resistant coating shall be applied after the equipment has been cleaned to remove such contaminants as corrosion products, dust, fingerprints, lubricating oils, mold release agents, sand, and solder fluxes. When tape is used on parts of equipment that are masked against treatments, the tape shall be an electrical grade. In the application of the MFP coating, care shall be taken to assure that the coating does not affect circuit constants and thereby degrade performance.

* 3.20.6 Fungus. - The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.18. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.21 Breakdown voltage, insulation. - Electrical insulation shall be capable of withstanding 1250 volts between primary and secondary position and 500 volts between primary or secondary and ground. After 10 second application of the voltage, there shall be no visible deterioration and the equipment shall meet specified performance requirements. (See 4.8)

3.22 Input. - Charger, Battery, PP-1660( )/G, shall be capable of operating from a 115 ± 10 volt, 60 ± 5% hertz, single phase AC power source. (See 4.14)

3.23 Output. - The output of the battery charger, under continuous operation in the ambient temperature specified, shall be as follows:

75 amperes DC at 8 volts DC
75 amperes DC at 16 volts DC
38 amperes DC at 32 volts DC

3.24 Efficiency. - The power required for operation on 115 volts, 60 hertz AC current, shall not exceed 2000 watts under continuous operation at rated output of 38 amperes at 32 volts, in an ambient temperature of 95°F. (See 4.16)

* 3.25 Interchangeability. - Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454. (See 4.9)

3.26 Service conditions. - The equipment shall meet the following service conditions:

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* 3.26.1 Operation (general). - The equipment shall operate under environmental service operating conditions specified in 3.26.2 through 3.26.4 with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement and 6.7

* 3.26.2 Service conditions operation requirements. - Prior to, during and after testing as specified for each of the following service conditions, the equipment shall meet the requirements of 3.23. (See 4.14)

* 3.26.2.1 Low temperature. - See 4.11.1.

* 3.26.2.2 High temperature. - See 4.11.2.

* 3.26.2.3 Humidity (moisture-resistance). - See 4.10.

3.26.3 Vibration, internal. - The amplitude of vibration of any part, sub-assembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 hertz. This examination may be limited to that which can be observed through the vents on each end of the case. At the conclusion of the test, for internal vibration (4.13), there shall be no evidence of physical or mechanical damage to the unit that may in any way prevent it from meeting any of its operational requirements. The equipment shall meet the output requirement specified in 3.23.

3.26.4 Bounce. - The equipment shall be subjected to the bounce test of 4.7. Upon completion of the test there shall be no evidence of physical or mechanical damage to the unit that may in any way prevent it from meeting any of its operational requirements and the equipment shall meet the output requirements specified in 3.23.

* 3.27 Electromagnetic interference characteristics. - The equipment shall meet the electromagnetic interference requirements specified in MIL-STD-461, Table A-1-MIL-E-55301, Equipment Class TS.

3.28 Preconditioning. - The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6. (Also see 4.5)

3.29 Technical literature and running spare parts. - Technical literature and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2)

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3.30 Workmanship. - The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.4 Cables
3.5 Carrying Case
3.13 Controls
3.14 Size and weight
3.15 Cleaning
3.18 Riveting
3.19 Soldering
3.20 Tropicalization

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.21)

* 4.3 Preproduction inspection. - This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B and group C (see tables II, III and IV, respectively). The preproduction inspection shall be performed in the order shown in table I.

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* TABLE I Order of preproduction inspection

Inspection	Unit 1	Unit 2	Unit 3
Non-damaging			
Temperature			
Low	1		
High	2		
Temperature Rise	3		
Vibration		1	
Electromagnetic compatibility			1
Potentially damaging			
Humidity (moisture-resistance)	4		
Fungus			2
Bounce		2	

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where Required
Finish	3.16
Marking	3.17

* 4.5 Quality conformance inspection of equipment before preparation for delivery. - Each unit of each lot of equipment shall be subjected to bounce preconditioning (See 3.28) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of bounce preconditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (see 6.6). Test equipment for Government verification inspection shall be made available by the contractor.

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* 4.5.1 Group A inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of table II. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in table II. Group A inspection shall be performed in any order which is satisfactory to the Government.

TABLE II Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL	
			Major	Minor
Visual & mechanical	3.30	4.19	1.5DPHU*	6.5DPHU
Output	3.23	4.14	1.5DPHU	**
Timer operational	3.8	4.15	1.5DPHU	**

* Defects per hundred units.
** All electrical defects are considered Major.

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to table III and to special procedures for small-sample inspection of MIL-STD-105. Group B inspection shall normally be performed on inspection lot that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. The AQL shall be 6.5 percent defective and the inspection level shall be S-4 for normal, tightened and reduced inspection.

TABLE III - Group B inspection

Inspection	Req. Para.	Insp. Para.
Dielectric strength	3.21	4.8
Efficiency	3.24	4.16
Interchangeability	3.25	4.9

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4.5.3 Group C inspection. - This inspection shall consist of the tests specified in Table IV, and shall be performed on sample units that have been subjected to and met group A and group B inspection. Sample units shall be selected in accordance with paragraph 4.5.3.1.

* TABLE IV - Group C inspection

Inspection	Req. Para.	Insp. Para.
Group I		
Low Temperature	3.26.2.1	4.11.1
High Temperature	3.26.2.2	4.11.2
Temperature rise	3.6, 3.7, 3.11	4.12
Humidity	3.26.2.3	4.10
Group II		
Bounce	3.26.4	4.7
Vibration	3.26.3	4.13
Group III (See 4.5.3.1.3)		
Electromagnetic Compatibility	3.27	4.17
Fungus and corrosion	3.20	4.18

4.5.3.1 Sampling for group C inspection. -

4.5.3.1.1 Group I - Table IV. - One sample shall be selected at random, from each 100 equipments produced or fraction thereof, and subjected to each inspection specified. The first sample shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

4.5.3.1.2 Group II - Table IV. - One sample shall be selected at random, from each 200 equipments produced or fraction thereof, and subjected to each inspection specified. The first sample shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

4.5.3.1.3 Group III - Table IV. - This inspection shall be required only when preproduction samples are not specified by contract. When required, one sample shall be selected at random at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3) No subsequent inspection for the test in this group will be required.

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4.5.3.2 Noncompliance. - The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials, as applicable, and (2) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and groups A and B inspection may be reinstated; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of conforming Group B and Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.6 Bounce preconditioning. - The unit shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch-thick strip of material under one corner or edge of the unit to a distahce of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, cleaned, or otherwise changed prior to subjection to inspection.

4.7 Bounce test: class F units (see 3.26.4). - The equipment shall be tested on the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

(a) Cover the tester bed with a panel of 1/2-inch plywood, with the grain, parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

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(b) Place the equipment on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and aft motion of the equipment against the back-stop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of 1-1/2 hours at 284 ± 2 rpm. Turn the equipment at the end of each 30 minutes so it will rest on each side and bottom.

4.8 Dielectric strength. - Charger, Battery PP-1660( )/G shall be tested to determine compliance with the requirement specified in 3.21. Rectifier stacks shall be disconnected for this test.

4.9 Inspection for dimensional interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.25. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

* 4.10 Humidity (moisture-resistance) test. - The equipment shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.26.2.3. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

* 4.11 Temperature test. -

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* 4.11.1 Low Temperature. - The equipment shall be tested in accordance with Method 502, Procedure 1, of MIL-STD-810. The storage temperature shall be -70°F, and the operating temperature shall be - 65°F. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.26.2.1.

* 4.11.2 High temperature. - The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be + 125°F. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.26.2.2.

4.12 Temperature rise. - The equipment shall be operated continuously under full load for a period of four hours (the first hour using fast charge and the remaining time on slow). The temperature rise of the rectifier, transformer and fan motor shall be determined using a standard laboratory thermo-couples and shall meet the requirements specified in paragraphs 3.6, 3.7 and 3.11.

* 4.13 Test for internal vibration. - Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.26.3.

(a) Secure the equipment directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, sub-assemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 hertz. The total excursion of the applied vibration shall be not less than 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one hertz per second and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.14 Output. - Charger, Battery, PP-1660( )/G shall be connected to single phase AC power source which can be varied from 105 to 125 volts. Using standard laboratory DC voltmeter and ammeter, the output shall be tested to determine compliance with the requirement specified in 3.23.

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4.15 Timer operational. - With the equipment connected to the specified power source (see 3.22), the timer shall be set at 15 minutes, and the output current shall be measured. The output current shall be measured again after the timer has reached zero to determine compliance with the requirement specified in 3.8.

4.16 Efficiency (see 3.24). - The equipment shall be tested as follows:

4.16.1 Test conditions. -

(a) Input 115 v, 60 hertz. (See 3.22)

(b) Ambient temperature 95°F.

(c) Battery voltage switch set at 24 volts.

(d) Wattmeter connected to the input.

4.16.2 Test method. - The input watts shall be measured while the output of the equipment is 38 amperes at 32 volts.

4.17 Electromagnetic compatibility. - The preproduction model, equipped for the reduction of radio interference in accordance with 3.27 shall be tested by the contractor or a commercial testing organization approved by the Government. The contractor shall furnish the contracting officer, for approval, an Electromagnetic Compatibility test report. The test report will include a block diagram of the test set up, a description of the interference reduction system applied, and a list of the test equipment utilized. Upon approval by the contracting officer, and provided all other requirements of the specification are met, the preproduction model shall be used as a visual standard for all production units.

* 4.18 Fungus test. - The equipment (including accessories) shall be tested in accordance with Method 508, Procedure I, of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation is not required. (See 3.20.6)

4.19 Visual and mechanical inspection. - Equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.20 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A, or C as specified.

5.1.1 Level A. -

5.1.1.1 Cleaning. - Charger, Battery PP-1660( )/G shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Drying shall be accomplished in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Unit packaging. - Unit packaging shall be in accordance with the applicable procedures of the methods prescribed in MIL-P-116 and as specified herein.

5.1.1.3.1 Technical literature. - Each technical literature shall be packaged method IC-1.

5.1.1.3.2 Charger, Battery PP-1660( )/G. - Each battery charger shall be packaged method III, as follows: Fabricate a wood and plywood blocking frame, the frame shall be designed to bridge and protect the protruding knobs and the meter face. Secure the blocking frame in place with water-resistant pressure-sensitive tape, conforming to PPP-T-76. Cushion the battery charger on all surfaces with cells and pads, fabricated of fiberboard conforming to PPP-F-320. Place the cushioned case within a close-fitting fiberboard box conforming to PPP-B-636 grade W5c. Close the box in accordance with the requirements of the box specification. In addition, seal all seams and joints with water resistant tape, not less than 2 inches wide, conforming to PPP-T-76.

5.1.2 Level C. - Charger, Battery, PP-1660( )/G, shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during direct shipment from the supply source to the first receiving activity.

5.2 Packing. - Packing shall be level A, B or C, as specified. (See 6.2(b))

5.2.1 Level A. -

5.2.1.1 Palletized load. - Charger, Battery PP-1660( )/G, packaged as specified in 5.1, a quantity of chargers shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accomodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s. The load shall be bonded to the pallet by strapping.

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5.2.1.2 Less than pallet load. - When quantities per destination, of battery charger, are less than a pallet load, a quantity of battery chargers, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, grade A, style E or MIL-B-43291. Closure shall be made in accordance with the appendix of the applicable box specification. Reinforcing shall be made by non-metallic strapping or pressure-sensitive tape banding conforming to PPP-S-760, type II or PPP-T-97, type IV, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B. - Charger, Battery PP-1660( )/G, packaged as specified in 5.1, shall be packed as specified in 5.2.1 except that fiberboard caps shall be class domestic and fiberboard boxes limited to PPP-B-640, class domestic.

5.2.3 Level C. - Charger, Battery PP-1660( )/G, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except that fiberboard caps, pallets and fiberboard boxes shall be of the type, size and kind commonly used for the purpose and shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use. - Charger, Battery, PP-1660( )/G is used to charge 6, 12 or 24 volt vehicular storage batteries. The equipment may be used, also, as a booster to crank the engine of a military vehicle when the vehicle's battery is in an advanced condition of discharge.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

(c) Preproduction inspection:

Preproduction pack(s) as follows:

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Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

(d) Marking and shipping of samples.

(e) Place of final inspection.

(f) Technical literature required. (See 3.29)

(g) Quantity of tools and running spare parts required. (See 3.29)

(h) Submission of the statement of treatment referenced in 3.20.1.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer unless otherwise specified in the contract.

6.3 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: PP-1660W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.4 Location of operational inspection. - It is desirable that the timer operational inspection (4.15) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.5 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.6 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

* 6.7 Qualitative materiel requirements. - The following service conditions were considered in the design of the equipment and are given here as a guide for the contractor where he is required to make a selection of parts, materials, processes, construction methods, etc.

* 6.7.1 Temperature. -

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(a) Operating: Ambient temperature in the range of 125°F to - 65°F. (The high temperature includes effect of sun-load). Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 6 hours, at any one time.

(b) Nonoperating: Exposure in the range of + 160°F to - 70°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 6 hours, at any one time.

6.8 Identification of changes. - The paragraphs and tables of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issues were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the asterisks and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-55532A(EL), CHARGER, BATTERY PP-1660( )/G

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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